

Neutrophil membrane-coated nanoparticles inhibit synovial inflammation and alleviate joint damage in inflammatory arthritis

Qiangzhe Zhang, Diana Dehaini, Yue Zhang, Julia Zhou, Xiangyu Chen, Lifen Zhang, Ronnie H. Fang, Weiwei Gao and Liangfang Zhang *

Rheumatoid arthritis is a common chronic inflammatory disorder and a major cause of disability. Despite the progress made with recent clinical use of anti-cytokine biologics, the response rate of rheumatoid arthritis treatment remains unsatisfactory, owing largely to the complexity of cytokine interactions and the multiplicity of cytokine targets. Here, we show a nanoparticle-based broad-spectrum anti-inflammatory strategy for rheumatoid arthritis management. By fusing neutrophil membrane onto polymeric cores, we prepare neutrophil membrane-coated nanoparticles that inherit the antigenic exterior and associated membrane functions of the source cells, which makes them ideal decoys of neutrophil-targeted biological molecules. It is shown that these nanoparticles can neutralize proinflammatory cytokines, suppress synovial inflammation, target deep into the cartilage matrix, and provide strong chondroprotection against joint damage. In a mouse model of collagen-induced arthritis and a human transgenic mouse model of arthritis, the neutrophil membrane-coated nanoparticles show significant therapeutic efficacy by ameliorating joint damage and suppressing overall arthritis severity.

Rheumatoid arthritis is a widespread and devastating autoimmune disease characterized by systemic inflammation that causes progressive joint damage and disability^{1,2}. The precise cause of rheumatoid arthritis remains elusive, and current treatment primarily targets the inflammatory response³. Though there has been some success, especially with the use of biologics that inhibit tumour necrosis factor alpha (TNF- α) and interleukin (IL)-1, existing approaches carry considerable limitations^{4,5}. Pathological inflammation in rheumatoid arthritis is orchestrated by a large number of molecules^{6,7}, and inhibition of one or a few may not be sufficient to halt or reverse disease progression^{8,9}. Under current treatment regimens, it remains poorly controlled in up to 30% of patients, and only a minor proportion of patients reach sustained clinical remission^{10,11}. Moreover, due largely to the multiplicity of cytokine targets, the toxicity of cytokine inhibition remains highly unpredictable, causing substantial adverse effects and safety concerns¹². Therefore, alternative anti-inflammatory approaches that overcome the complexity and heterogeneity of the inflammatory network are highly desirable for effective rheumatoid arthritis treatment¹³.

Recently, cell-membrane-coated nanoparticles have emerged as a promising therapeutic platform^{14,15}. Made by fusing natural cell membranes onto synthetic cores, these nanoparticles inherit the antigenic profile of the source cells, enabling them to act as decoys that can absorb and neutralize heterogeneous and complex pathological molecules, regardless of their structural specificity¹⁴. For example, nanoparticles coated with a red blood cell membrane (RBC-NPs) have demonstrated the ability to bind bacterial pore-forming toxins¹⁴ and pathological autoantibodies¹⁶, diverting them away from and preventing their attack of healthy RBCs. Furthermore, macrophage membrane-coated nanoparticles have shown the ability to bind and neutralize endotoxins that would otherwise trigger immune activation¹⁷.

The advancement of cell-mimicking nanoparticles in biodegradation inspired us to develop these nanoparticles as an anti-inflammatory strategy to address the aforementioned challenges facing current treatment. Joint inflammation and damage in rheumatoid arthritis are mediated by the influx of an immune cell mixture into the synovial joint space¹⁸. Among them, neutrophils play an important role because of their actions in resolving inflammation and repairing tissues damage¹⁹. Neutrophils have been found to produce microvesicles that enter cartilage and protect the joint in inflammatory arthritis^{20,21}. Moreover, neutrophils are also responsible for initiating and perpetuating rheumatoid arthritis progression²². In rheumatoid arthritis, various chemoattractants have been identified that promote neutrophil migration into the joints²³. Subsequent neutrophil activation stimulates synovial cells to produce chemokines that amplify neutrophil recruitment²⁴. Neutrophil activation and granular content release contribute directly to cartilage destruction and bone resorption²⁵. Importantly, remission and subsequent reversion of rheumatoid arthritis have been linked with a decrease in neutrophil recruitment to the synovial fluid²⁶. Here, we have developed neutrophil-like nanoparticles (denoted 'neutrophil-NPs') and investigated their use as a broad-spectrum anti-inflammatory agent for rheumatoid arthritis management. By displaying neutrophil plasma membrane on their surface, the neutrophil-NPs were anticipated to mimic the source cells and thus to bind with immunoregulatory molecules that would otherwise target endogenous neutrophils (Fig. 1a).

Nanoparticle preparation and characterization

To synthesize neutrophil-NPs, plasma membrane derived from purified and activated human peripheral blood neutrophils (Supplementary Fig. 1) was coated onto poly(lactic-co-glycolic acid) (PLGA) polymeric cores. The completeness of membrane coating was verified, which provided neutrophil-NPs with superior

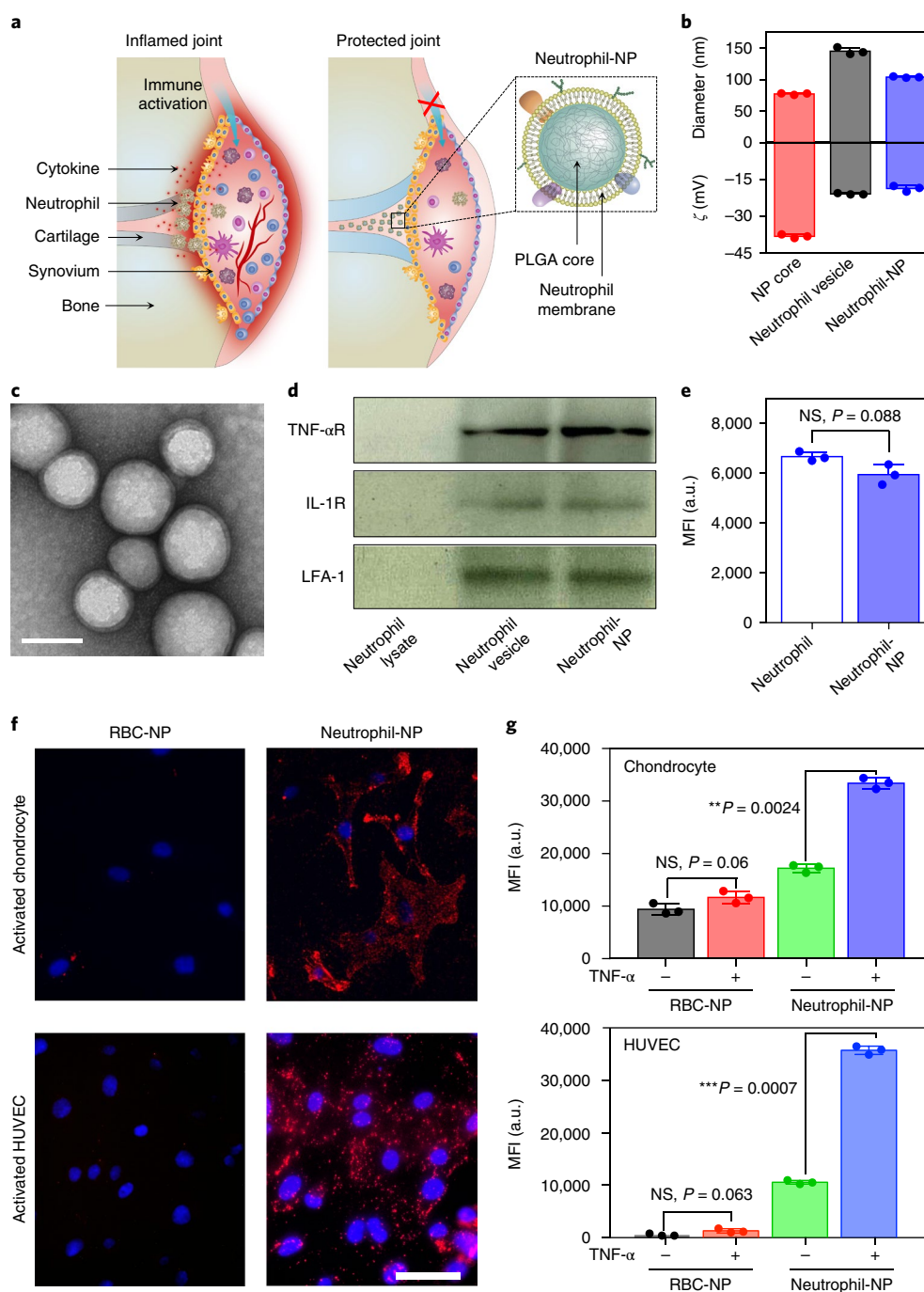


Fig. 1 | Preparation and characterization of neutrophil-NPs. **a**, Schematic representation of neutrophil-NPs designed for suppressing synovial inflammation and ameliorating joint destruction in inflammatory arthritis. Neutrophil-NPs are constructed by wrapping polymeric cores in natural human neutrophil membranes, which mimic source cells to bind with immunoregulatory molecules without potentiating the immune cascades for disease progression. **b**, DLS measurements of neutrophil-NP hydrodynamic size (diameter) and zeta potential (ζ). **c**, Representative image of neutrophil-NPs examined with transmission electron microscopy. Samples were stained with uranyl acetate. Scale bar, 100 nm. **d**, Characteristic protein bands of neutrophil cell lysates, neutrophil membrane-derived vesicles and neutrophil-NPs resolved by western blotting. **e**, Comparison of fluorescence intensity measured in neutrophils ($\sim 2.5 \times 10^6$ cells) and neutrophil-NPs (100 μ l of suspensions, 0.2 mg ml $^{-1}$ protein content) stained with allophycocyanin anti-mouse LFA antibodies. NS, not significant. **f**, Fluorescent images of chondrocytes and human umbilical vein endothelial cells (HUVECs) after incubation with neutrophil-NPs or RBC-NPs. Red represents nanoparticles and blue represents nuclei. Cells were activated with TNF- α before nanoparticle incubation. Scale bar, 50 μ m. **g**, Flow cytometric analysis of nanoparticle binding to human chondrocytes or HUVECs ($n = 3$). Statistical analysis was performed using paired two-tailed t -test. Data presented as mean $\pm$ s.d. In all datasets, $n = 3$ independent experiments using the same batch of human neutrophil-NPs.

colloidal stability (Supplementary Fig. 2). Dynamic light scattering (DLS) measurements revealed that the hydrodynamic diameter of the neutrophil-NPs increased by ~ 18 nm compared with that of the uncoated PLGA cores; the surface zeta potential was less

negative than the cores but comparable to neutrophil membrane-derived vesicles (Fig. 1b). The neutrophil-NPs were visualized by transmission electron microscopy after uranyl acetate staining and showed a spherical core-shell structure (Fig. 1c), consistent with a

unilamellar membrane coating around the polymeric core^{14,27}. Furthermore, immunoblotting confirmed the presence and enrichment of key surface antigens, including TNF- α receptor (TNF- α R), IL-1 receptor (IL-1R) and lymphocyte function-associated antigen 1 (LFA-1) on neutrophil-NPs, further confirming the translocation of neutrophil membrane onto the polymeric cores (Fig. 1d). A right-side-out protein orientation was verified by immunostaining and quantifying LFA-1 antigen on neutrophil-NPs and comparing with that on neutrophil cells with an equal amount of membrane proteins (Fig. 1e). Overall, a series of quality assurance specifications for the production of neutrophil-NPs was established to ensure physicochemical and biological reproducibility of the nanoparticles (Supplementary Table 1).

In rheumatoid arthritis, neutrophils are known to engage in receptor-mediated adhesion with cytokine-activated chondrocytes²⁸. Here, human neutrophil-NPs were fluorescently labelled and added to monolayers of chondrocytes activated with TNF- α and HUVECs. RBC-NPs were tested as a control because they had similar particulate structures as neutrophil-NPs, but RBCs are less plastic in promoting and resolving inflammations compared to neutrophils²⁹. After incubation and washing, significant fluorescence was observed on cells incubated with neutrophil-NPs, but not with RBC-NPs (Fig. 1f). Flow cytometry measurements confirmed that activated chondrocytes and HUVECs, when incubated with neutrophil-NPs, showed a significant increase in mean fluorescence intensity (MFI) compared to naive cells (without TNF- α activation) (Fig. 1g). These results demonstrate the ability of neutrophil-NPs to target inflamed cells conferred by their neutrophil membrane coating. The binding is probably attributed to specific interactions between LFA-1 on the neutrophil membrane and the intercellular activation molecule 1 (ICAM-1) overexpressed on activated chondrocytes and HUVECs^{22,23}.

Inhibition of pro-arthritis factors

We first tested the binding capability of human neutrophil-NPs to IL-1 β and TNF- α , which play prominent roles in initiating rheumatoid arthritis and promoting disease progression^{6,8}. Based on the measured binding kinetic profiles, it was determined using the Hill equation that neutrophil-NPs had an IC₅₀ (half maximal inhibitory concentration) value of 188 $\mu\text{g ml}^{-1}$ for IL-1 β binding and 1,327 $\mu\text{g ml}^{-1}$ for TNF- α binding (Fig. 2a and Supplementary Fig. 3). We then evaluated the inhibition of chondrocyte activation induced by IL-1 β and TNF- α . After 6 h of incubation with 10 ng ml⁻¹ IL-1 β or 100 ng ml⁻¹ TNF- α , significant chondrocyte activation was observed, indicated by an increased level of ICAM-1 expression (Supplementary Fig. 4a). However, expression levels decreased with an increase in neutrophil-NPs added to the medium, suggesting a dose-dependent inhibition effect (Fig. 2b and Supplementary Fig. 4b). An IC₅₀ value of 0.88 mg ml⁻¹ for inhibiting IL-1 β and 0.44 mg ml⁻¹ for TNF- α were obtained. We also evaluated the capability of neutrophil-NPs to inhibit chondrocyte apoptosis caused by cytokines. In the study, 48 h incubation of IL-1 β or TNF- α with chondrocytes resulted in 80% of apoptotic cells (Supplementary Fig. 5a), but the fraction of apoptotic cells decreased with an increase in neutrophil-NPs (Fig. 2c and Supplementary Fig. 5b). IC₅₀ values of 1.58 mg ml⁻¹ for inhibiting IL-1 β and 0.98 mg ml⁻¹ for inhibiting TNF- α were observed. Neutrophil-NPs also inhibited HUVEC activation in a similar dose-dependent manner (Supplementary Fig. 6). In contrast, RBC-NPs did not show any detectable cytokine neutralization effects (Supplementary Fig. 7).

The ability of human neutrophil-NPs to inhibit pro-arthritis factors was further tested with human synovial fluid (hSF) samples obtained from three rheumatoid arthritis patients. Chondrocytes showed enhanced activation after 6 h of incubation with all three hSF samples, with 2.0-, 2.4- and 2.2-fold increases, respectively. In contrast, when the hSF samples were pre-treated

with neutrophil-NPs at a concentration of 2 mg ml⁻¹ for 2 h, they showed significantly reduced activation of chondrocytes with 1.5-, 1.5- and 1.4-fold increases, respectively (Fig. 2d and Supplementary Fig. 8a). Meanwhile, prolonged incubation with hSF samples caused 53.5, 67.3 and 53.6% chondrocyte apoptosis, while pre-treatment with neutrophil-NPs reduced these values to 5.3, 15.4 and 27.2%, respectively (Fig. 2e and Supplementary Fig. 8b). Neutrophil-NPs also inhibited HUVEC activation by all three hSF samples (Supplementary Fig. 8c). Moreover, the presence of neutrophil-NPs induced a nearly twofold decrease of MMP-3 plateau concentration in the culture medium for all three samples (Fig. 2f). Neutrophil-NPs also significantly increased the production of aggrecan from hSF-activated chondrocytes (Fig. 2g). Quantitatively, the three hSF-activated chondrocyte samples had aggrecan production levels of 65.8, 55.4 and 61.1%, respectively, compared to naive chondrocytes, which were increased to 85.9, 82.8 and 98.6%, respectively, in the presence of neutrophil-NPs (Fig. 2h). Neutrophil-NPs reduced aggrecanase caused by either aggrecanase or MMP, attributable to their neutralization of pro-arthritis cytokines (Supplementary Fig. 9). It should be noted that different hSF samples caused different levels of cell activation, apoptosis and catabolic status, probably due to the variation of pro-arthritis profiles of different individuals³⁰. Nevertheless, neutrophil-NPs showed varying but significant neutralization efficacy across all samples without pre-existing knowledge of the hSF composition. Further study found that the catabolic status of chondrocytes remained unaltered when neutrophil-NPs alone were used without hSF, suggesting that the observed effects of neutrophil-NPs on chondrocytes were indeed related to their neutralization of pro-arthritis factors (Supplementary Fig. 10).

Cartilage penetration and chondroprotection

To study the ability of neutrophil-NPs to penetrate injured cartilage in a mouse model, mouse neutrophil-NPs were fabricated, which showed properties comparable to those of human neutrophil-NPs (Supplementary Figs. 11–14 and Supplementary Table 2). Mouse femoral head explants were collected and cultured in medium supplemented with IL-1 β to mimic inflammatory arthritis²¹. Fluorescently labelled mouse neutrophil-NPs or RBC-NPs were added to the explants, followed by 18 h of incubation before sectioning. Using fluorescence microscopy, we observed the accumulation of neutrophil-NPs on the distal region of the femoral heads, with clear penetration into the tissue (Fig. 3a). Dual fluorescence labelling demonstrated co-localization of the polymeric cores and the neutrophil membranes during cartilage penetration (Supplementary Fig. 15). Below the surface zone, neutrophil-NPs distributed throughout the cartilage matrix. A strong nanoparticle signal in close proximity to the stained nuclei implied intracellular uptake by stimulated chondrocytes (Supplementary Fig. 16). RBC-NPs also showed accumulation in the distal surface region, but with much weaker fluorescence intensity (Fig. 3b). Fluorescence intensity at different penetration depths was quantified by sampling image stacks with a width of 5 μm and normalized to the outermost section of tissue treated with neutrophil-NPs. The signal of neutrophil-NPs decayed with increasing penetration depth, which dropped to 1% at a depth of 140 μm (Fig. 3c). In contrast, the signal of RBC-NPs was ~60% that of neutrophil-NPs at the distal surface and decayed rapidly to 1% at a depth of 30 μm . The enhanced cartilage penetration of neutrophil-NPs is probably attributable to the adhesion interactions between neutrophils and chondrocytes. Besides LFA-1 and ICAM-1, neutrophils may exploit other ligands for adhesion^{31,32}, but the top-down fabrication of neutrophil-NPs can bypass the need for ligand identification and conjugation^{14,15}.

We next examined whether neutrophil-NPs provide chondroprotection in inflammation-induced cartilage damage. We extracted the femoral heads from mice and maintained them in culture medium

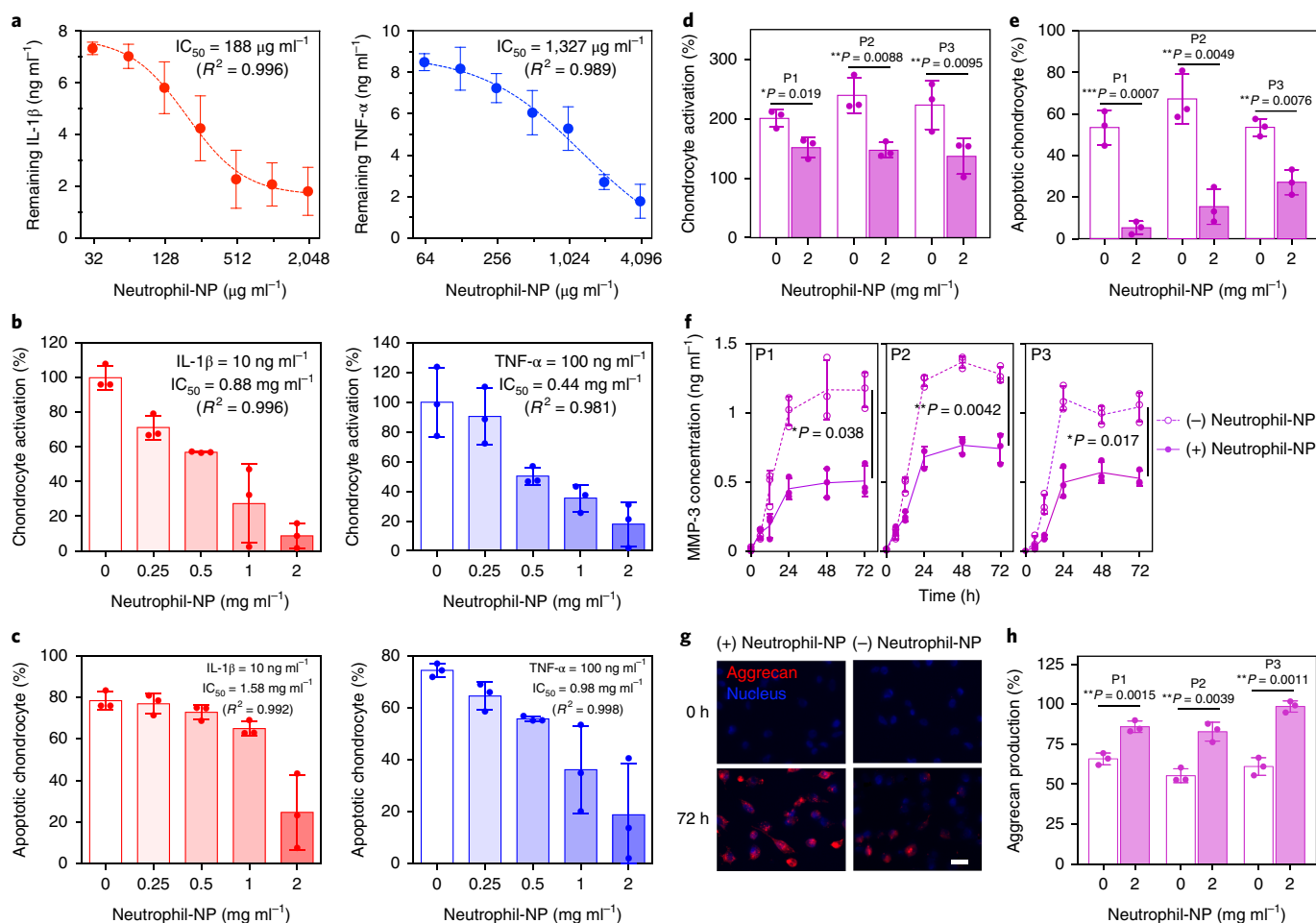


Fig. 2 | Neutrophil-NPs inhibit pro-arthritis factors in vitro. **a**, Binding capacity of human neutrophil-NPs with IL-1 β (left) and TNF- α (right). **b,c**, Dose-response inhibition of chondrocyte activation (**b**) and chondrocyte apoptosis (**c**), induced by IL-1 β (left) or TNF- α (right). Concentration of IL-1 β was maintained at 10 ng ml $^{-1}$ and that of TNF- α at 100 ng ml $^{-1}$; neutrophil-NP concentrations were varied from 0 to 2 mg ml $^{-1}$. Values measured from chondrocytes treated with cytokines only (without neutrophil-NPs) served as 100%. **d-f**, Effects of neutrophil-NPs on chondrocyte activation (**d**), apoptosis (**e**) and matrix metalloproteinase-3 (MMP-3) secretion (**f**), elicited by human synovial fluid (hSF) samples collected from three rheumatoid arthritis patients. In the study, chondrocyte activation was quantified by measuring the level of ICAM-1 expression. Chondrocyte apoptosis was measured by examining mitochondrial activity through cell staining with 3,3'-dihexyloxycarbocyanine iodide (DiOC $_6$) dye. Neutrophil-NPs were removed following incubation with the cytokine solutions or hSF samples. **g,h**, Effects of neutrophil-NPs on aggregate production from synovial fluid-activated chondrocytes. Representative fluorescence images (**g**; scale bar, 20 μ m) and quantification of aggregate based on fluorescence staining (**h**) from hSF-activated chondrocytes with or without neutrophil-NPs. Values measured from naive chondrocytes without addition of hSF nor neutrophil-NPs served as 100%. Statistical analysis for hSF-induced chondrocyte activation, apoptosis and aggrecan production was performed using paired *t*-test. Analysis for chondrocyte MMP-3 secretion was performed using repeated-measure one-way analysis of variance (ANOVA). Data presented as mean $\pm$ s.d. **P* $\leq$ 0.05, ***P* $\leq$ 0.01, ****P* $\leq$ 0.001. In all datasets, *n* = 3 independent experiments using the same batch of human neutrophil-NPs. IC_{50} values were derived from the variable slope model using Graphpad Prism 7.

under various conditions, followed by safranin-O staining for cartilage histopathology analysis²¹. The control femoral head sample without additional manipulation showed cartilage with an intact surface and perichondrium with a homogeneous cartilage matrix (Fig. 3d). When the culture medium was supplemented with 10 ng ml $^{-1}$ IL-1 β , the femoral head showed obvious cartilage loss with matrix fibrillation that extended vertically downward into the mid zone. The sample also showed significant cartilage matrix loss in both superficial zones and fissured domains (Fig. 3e). When RBC-NPs were added, the stained section showed a pathology similar to the sample treated with IL-1 β alone (Fig. 3f). However, when neutrophil-NPs were added, cartilage damage induced by IL-1 β was halted. The sample showed a mostly intact surface with a large area of continuous cartilage matrix preserved below the superficial zone with reduced areas of disorganization, increased number of chondrocytes

and absence of fissures (Fig. 3g). The cartilage content, represented by areas with safranin-O staining (dark red), was further quantified (Fig. 3h). In tissues cultured with IL-1 β alone or supplemented with RBC-NPs, only about 40% of the cartilage remained when compared to the unmanipulated control. In contrast, the tissue cultured with IL-1 β and neutrophil-NPs was able to maintain more than 80% of its cartilage. Notably, without adding cytokines, neutrophil-NPs or RBC-NPs alone did not alter the cartilage content, implying that the observed chondroprotective effect was mediated by neutralization of arthritis factors (Supplementary Fig. 17).

Therapeutic and prophylactic efficacies

The efficacy of neutrophil-NPs to ameliorate joint destruction was first evaluated using a murine model of collagen-induced arthritis (CIA)³³. Following arthritis induction, CIA mice developed mild

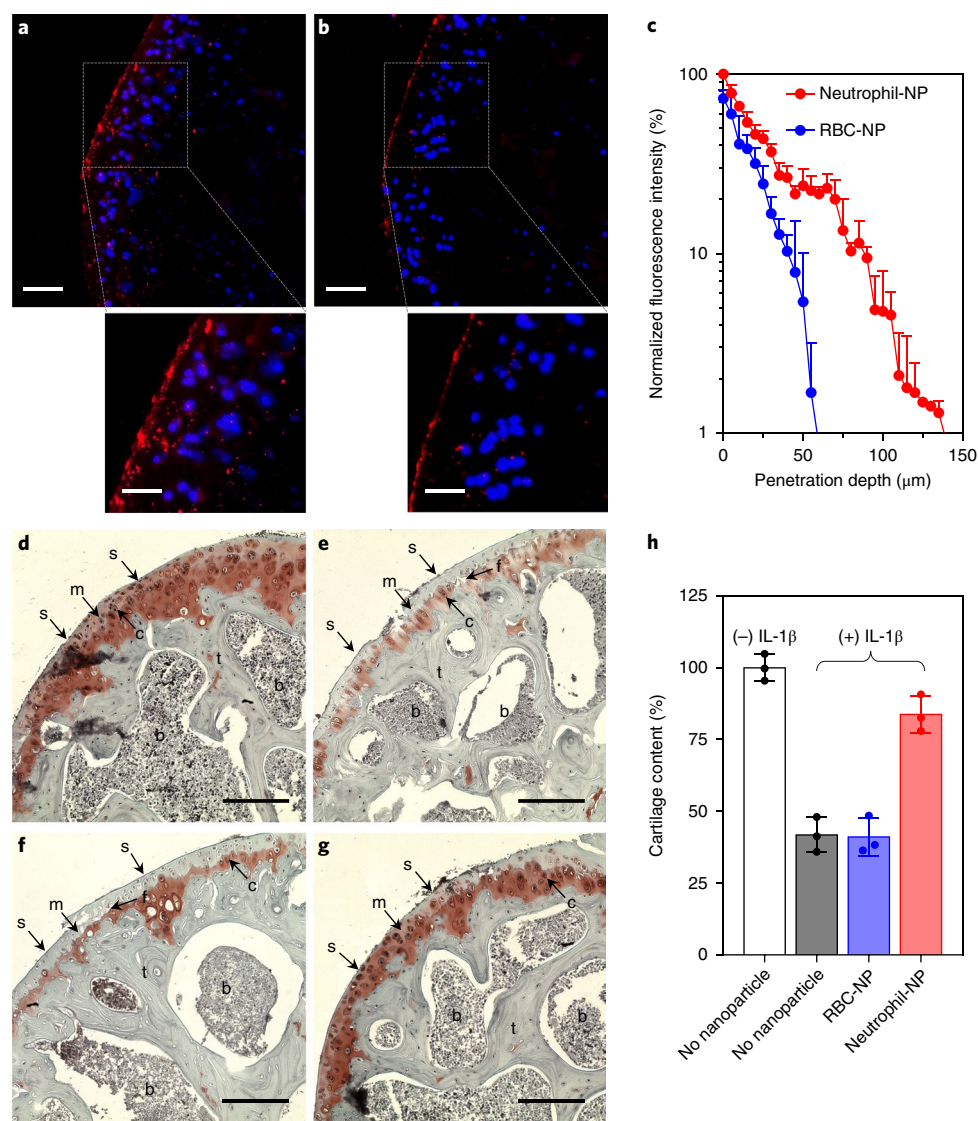


Fig. 3 | Neutrophil-NPs enhance cartilage penetration and confer chondroprotection. **a, b**, Representative fluorescence images of a cross-section of mouse femoral heads stimulated with recombinant mouse IL-1 β and incubated with fluorescence-labelled mouse neutrophil-NPs (**a**) or RBC-NPs (**b**). Scale bars, 100 μm . **c**, Quantitative analysis of nanoparticle penetration depth into IL-1 β -stimulated mouse femoral heads. **d–g**, Representative safranin-O-stained cross-sections of mouse femoral head explants under various treatment conditions, including without either IL-1 β stimulation or nanoparticle treatment (**d**), stimulated but without nanoparticle treatment (**e**), stimulated and treated with RBC-NPs (**f**) and stimulated and treated with neutrophil-NPs (**g**). s, superficial zone; m, mid zone; c, chondrocytes; f, fissure; b, bone marrow; t, bone trabeculae. Scale bars, 100 μm . **h**, Quantitative analysis of safranin-O-positive area among the above groups. Data presented as mean $\pm$ s.d. In all datasets, $n=3$ femoral heads from different mice.

swelling of the knee joints where mouse neutrophil-NPs were injected (Fig. 4a). Mice injected with PBS, anti-IL-1 β or anti-TNF- α served as controls^{34,35}. At the study endpoint, the transverse knee diameter of mice injected with neutrophil-NPs was comparable to that of mice treated with anti-IL-1 β or anti-TNF- α , but significantly smaller than the PBS group, indicating a reduction in cellular influx and oedema formation (Fig. 4b). The knee joints of the mice were also sectioned for histological analysis. Haematoxylin and eosin (H&E)-stained sections from the neutrophil-NP group showed an even distribution of chondrocytes within the articular cartilage, without obvious degeneration. In contrast, chondrocytes in the PBS group were largely absent and an intense neutrophil infiltration was observable in the joints and synovium. Anti-cytokines-treated groups showed less neutrophil infiltration but slight enlargement and depletion of chondrocytes (Fig. 4c). Sections stained with safranin-O showed a higher level of sulfated glycosaminoglycans

(sGAGs) in the cartilage of the neutrophil-NP group than the PBS group, but similar to the anti-cytokine groups (Fig. 4d,e). The phenotype of fibroblast-like synoviocytes (FLSs) in the synovial intimal lining was also examined. Specifically, we stained FLSs for the expression of CD248 and fibronectin, known to upregulate IL-1 β and TNF- α ^{36,37}. Elevated expression of these markers indicates an aggressive phenotype that invades the extracellular matrix and exacerbates joint damage in rheumatoid arthritis³⁸. As shown in Fig. 4f, the neutrophil-NP group showed weak staining of CD248 with few positively stained cells in the synovium, while a large number of CD248-positive cells were found in sections of the PBS group. Similarly, fibronectin-positive cells were largely present in PBS control joints, but not within the synovium of neutrophil-NP and anti-cytokine groups.

We further examined the cartilage protection efficacy of neutrophil-NPs using a human transgenic mouse model of inflammatory

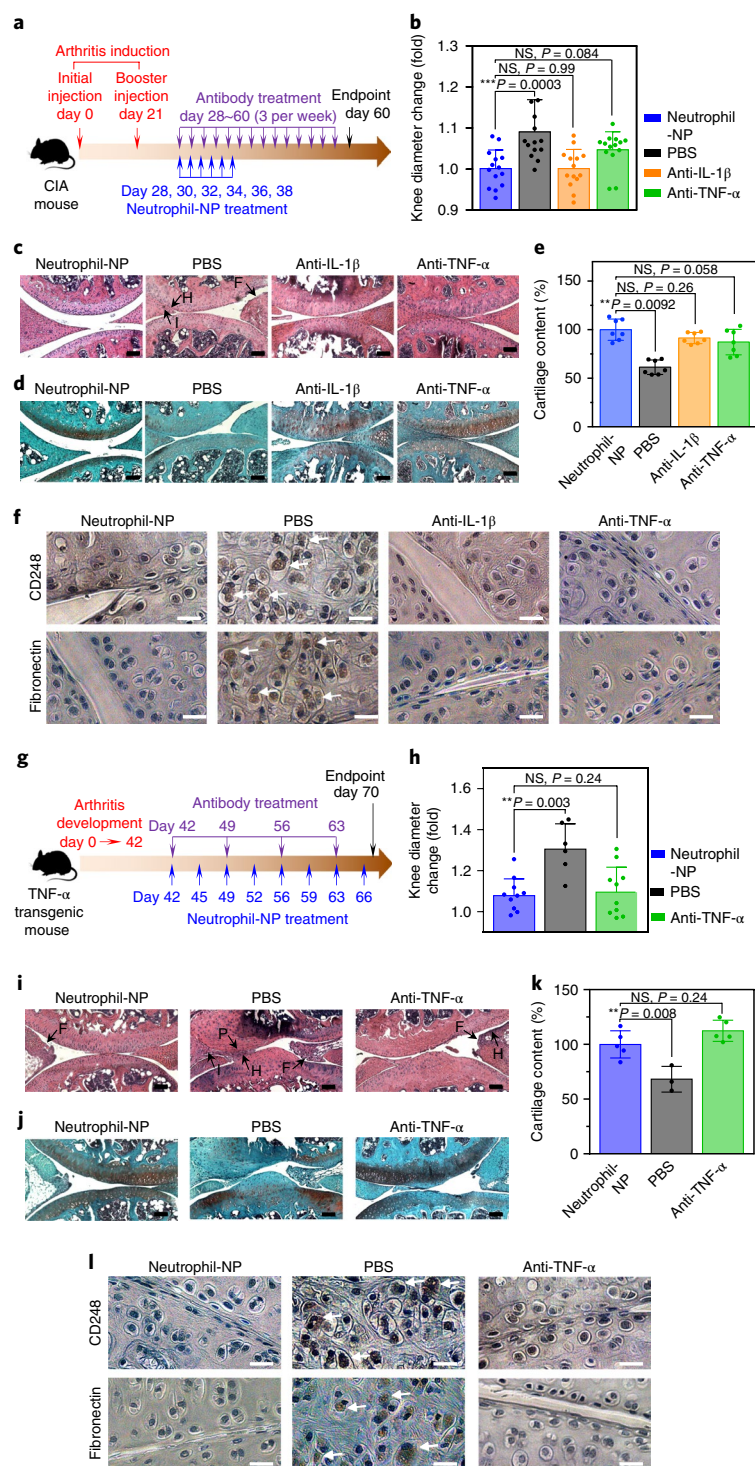


Fig. 4 | Neutrophil-NPs ameliorate joint destruction in a mouse model of collagen-induced arthritis and a human transgenic mouse model of inflammatory arthritis. **a**, The study protocol of a therapeutic regimen with a CIA mouse model. **b**, Change of hind knee diameter on day 60 after CIA induction compared to that on day 0. **c,d**, Representative images of H&E staining (**c**) and safranin-O staining (**d**) on knee sections from mice treated with neutrophil-NPs, PBS, anti-IL-1 β antibody or anti-TNF- α antibody. Scale bars, 100 μ m. F, synovial membrane fibrillation; H, synovium hyperplasia; I, immune cell infiltration. **e**, Cartilage content was quantified from safranin-O-stained sections of mice treated in different groups. **f**, Representative images of immunohistochemical staining for CD248 (top row) and fibronectin (bottom row) on knee sections from mice treated in different groups. Scale bars, 10 μ m. **g**, The study protocol of a therapeutic regimen with a human TNF- α transgenic mouse model of inflammatory arthritis. **h**, Change of hind knee diameter on day 70 compared to that on day 0. **i,j**, Representative images of H&E staining (**i**) and safranin-O staining (**j**) on knee sections from mice treated with neutrophil-NPs, PBS or anti-TNF- α antibody. Scale bars, 100 μ m. F, synovial membrane fibrillation; H, synovium hyperplasia; I, immune cell infiltration; P, pannus formation. **k**, Cartilage content was quantified from safranin-O-stained sections of mice treated in different groups. **l**, Representative images of immunohistochemical staining for CD248 (top row) and fibronectin (bottom row) on knee sections from mice treated in different groups. Scale bars, 10 μ m. Statistical analysis was performed using one-way ANOVA with Dunnett's post hoc analysis. Data presented as means $\pm$ s.d. In the CIA mouse study, $n = 7$ mice for all groups. In the transgenic mouse study, $n = 5$ mice for neutrophil-NP and anti-TNF- α antibody groups, $n = 3$ mice for PBS group. $*P \leq 0.05$, $**P \leq 0.01$, $***P \leq 0.001$.

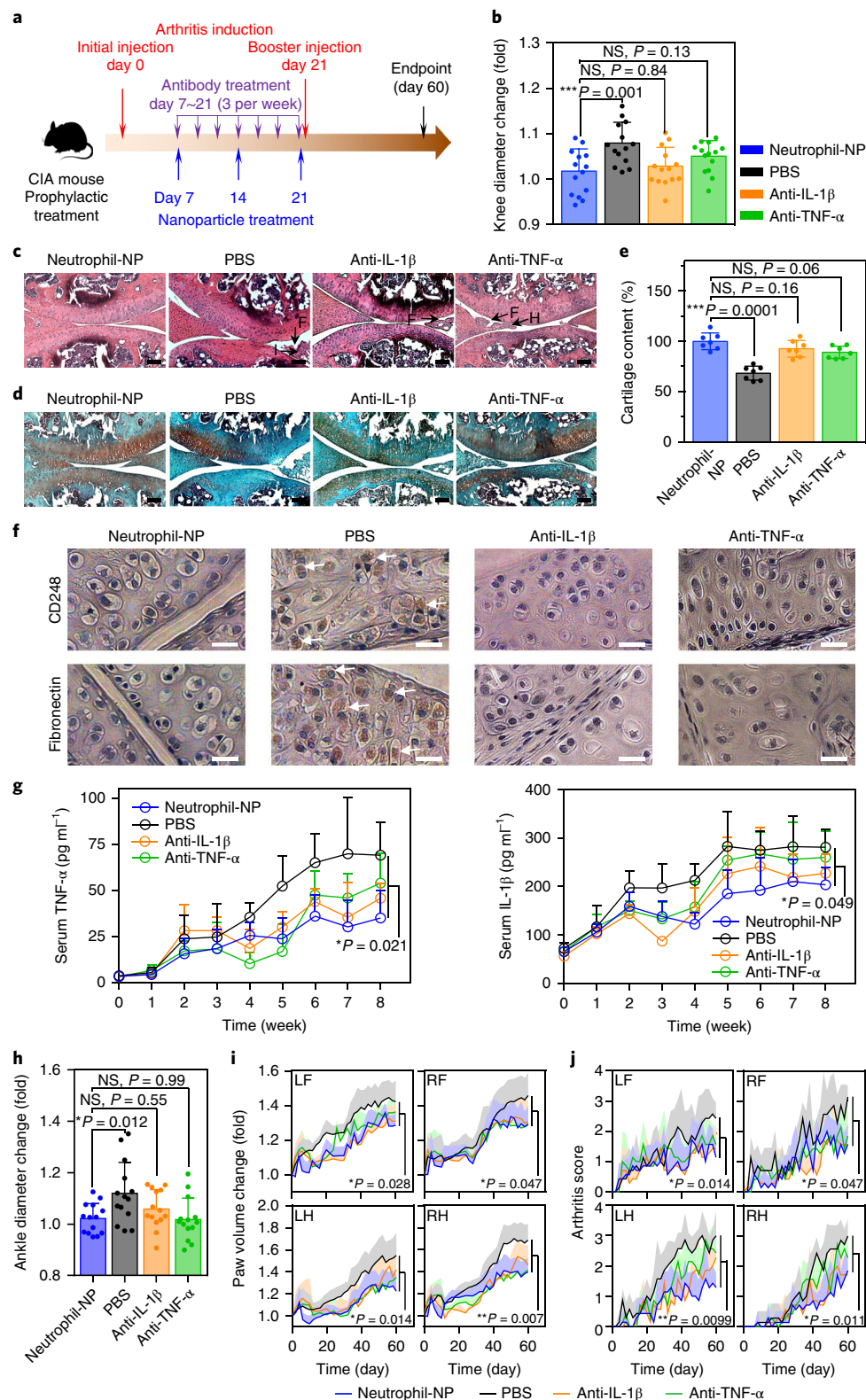


Fig. 5 | Neutrophil-NPs ameliorate joint destruction and elicit a systemic therapeutic response following a prophylactic regimen. **a**, The study protocol of a prophylactic regimen with a CIA mouse model. **b**, Change of hind knee diameter on day 60 after arthritis induction compared to that on day 0. On day 60, mice were euthanized and the hind knees were sectioned for histological analysis. **c, d**, Representative images of H&E staining (**c**) and safranin-O staining (**d**) on knee sections from mice treated with neutrophil-NPs, PBS, anti-IL-1β antibody or anti-TNF-α antibody. Scale bars, 100 μm. **f**, synovial membrane fibrillation; **H**, synovium hyperplasia; **I**, immune cell infiltration. **e**, Cartilage content quantified from safranin-O-stained sections of mice treated with neutrophil-NPs, PBS, anti-IL-1β antibody or anti-TNF-α. **f**, Representative images of CD248 and fibronectin immunohistochemical staining on knee sections. Scale bars, 10 μm. **g**, Concentration profiles of TNF-α and IL-1β in the serum of CIA mice treated with different groups. **h**, Change of hind ankle diameter on day 60 after arthritis induction compared to that on day 0. **i, j**, Values of paw volume (**i**) and arthritis score (**j**) recorded every other day for a total of 60 days. Statistical analysis was performed using one-way ANOVA with Dunnett's post hoc analysis. Arthritis score was analysed by the Kruskal-Wallis non-parametric test with Dunnett's post hoc analysis. All data points represent means $\pm$ s.d. ($n = 7$ CIA mice). $*P \leq 0.05$, $**P \leq 0.01$, $***P \leq 0.001$.

arthritis, where mice express a human TNF- α transgene and spontaneously develop arthritis³⁹. In the study, mouse neutrophil-NPs, anti-TNF- α or PBS were injected into transgenic mice after 6 weeks of arthritis development (Fig. 4g)⁴⁰. Similar to the CIA model, at the study endpoint the knee diameter of the neutrophil-NP group was smaller than that of the PBS group, but comparable with the anti-TNF- α group (Fig. 4h). H&E-stained sections from the neutrophil-NP group showed the preservation of chondrocytes in the articular cartilage, contrasting with the chondrocyte depletion and matrix deformation in sections of the PBS group, but consistent with features in sections of the anti-TNF- α group (Fig. 4i). In addition, the neutrophil-NP group maintained a higher level of sGAGs in the cartilage than the PBS group, but comparable to the anti-TNF- α group (Fig. 4j,k). Moreover, most FLSs in the neutrophil-NP and anti-TNF- α groups remained CD248-negative and fibronectin-negative, in contrast with the positive ones in the PBS group (Fig. 4l).

Finally, to evaluate the broad applicability of the neutrophil-NPs, we tested their anti-arthritis effectiveness using CIA mice with early-stage arthritis following a prophylactic regimen (Fig. 5a). We first observed chondral protection at the knee joints where the mouse neutrophil-NPs were injected. At the study endpoint the knee diameter measured from mice in the neutrophil-NP group was significantly smaller than that in the PBS group, but comparable with values in the anti-IL-1 β and anti-TNF- α groups (Fig. 5b). Histological analysis also confirmed a reduction of immune infiltration and preservation of cartilage in mice receiving neutrophil-NPs (Fig. 5c-e). In addition, most FLSs in the PBS group showed CD248-positive and fibronectin-positive phenotypes, as opposed to the negative ones in the neutrophil-NP and anti-cytokines groups (Fig. 5f). Notably, similar inhibition of CD248 and fibronectin expression with neutrophil-NPs was also observed in synovium tissues further away from cartilage (Supplementary Fig. 18). Although the neutrophil-NPs were injected locally and retained primarily at the knee joints (Supplementary Fig. 19), they might still be able to neutralize diffusive arthritogenic factors and thus elicit a systemic therapeutic response, thereby alleviating overall disease progression and severity. To examine the potential systemic response, we monitored the serum levels of TNF- α and IL-1 β in CIA mice, which are known to increase with the onset of arthritis and correlate strongly with disease severity. Indeed, alleviated concentrations of these cytokines were observed for the neutrophil-NP and anti-cytokines groups as compared to the PBS group, indicating effective reduction of arthritis at the systemic level (Fig. 5g). In addition, the neutrophil-NP group showed the smallest diameter of ankle below the knee (Fig. 5h). Meanwhile, paws of mice receiving PBS developed erythema and severe swelling; this effect was significantly lessened for mice treated with neutrophil-NPs or anti-cytokines, quantified by measuring paw volume (Fig. 5i and Supplementary Fig. 20) and paw cartilage loss (Supplementary Fig. 21). Moreover, blinded scoring of the swelling and redness of mouse paws was conducted to evaluate the severity of arthritis in the experimental mice⁴¹. The results showed that neutrophil-NP-treated mice had significantly lower arthritis score than the PBS group (Fig. 5j).

Conclusions

We have developed neutrophil-NPs and demonstrated their potential as an anti-inflammatory strategy for rheumatoid arthritis management. Unlike existing anti-cytokine agents that inhibit specific and limited targets, neutrophil-NPs provide a function-driven, broad-spectrum and disease-relevant blockade that dampens the inflammation cascade in the disease process. In addition, the direct use of membrane from the effector cells of the disease allows neutrophil-NPs to neutralize highly relevant inflammatory factors without the need to identify them. Even if the target is known, the top-down fabrication of neutrophil-NPs can avoid the necessity to engineer binding motifs that are often difficult to synthesize and conjugate.

Notably, the rheumatoid synovium contains a mixture of cells with origins of monocytes, macrophages, dendritic cells and T cells, which together orchestrate the rheumatoid arthritis process^{6,42}. The membrane of these cells can also be derived for coating and combined with other nanotherapeutic agents for targeted and synergistic therapy for rheumatoid arthritis^{43,44}. Similar to rheumatoid arthritis, various infections, traumatic injuries and chronic diseases are marked by inflammatory responses that can damage host tissue and cause organ dysfunction⁴⁵. The neutrophil-NPs developed here can probably be adopted to address some of these diseases, promising to eventually improve the clinical outcomes of patients. In this study, neutrophils were collected from human or mouse whole blood for membrane derivation (Supplementary Table 3). Regarding future clinical translation of neutrophil-NPs, it is critical to consider scale-up and manufacturing issues because of the use of natural cell membrane. In terms of the source materials, bioprocesses for ex vivo production of neutrophils at a clinical scale can supply a large quantity of neutrophil membrane⁴⁶. Meanwhile, significant progress has been made in producing human cells with universal immune compatibility, which would also enable cell supply for clinical studies⁴⁷. These technological advances together offer a promising prospect for the translation of neutrophil-NPs and cell membrane-coating technology in general.

Methods

Methods, including statements of data availability and any associated accession codes and references, are available at <https://doi.org/10.1038/s41565-018-0254-4>.

Received: 9 October 2017; Accepted: 2 August 2018;

Published online: 3 September 2018

References

- Smolen, J. S. & Aletaha, D. Rheumatoid arthritis therapy reappraisal: strategies, opportunities and challenges. *Nat. Rev. Rheumatol.* **11**, 276–289 (2015).
- Firestein, G. S. Evolving concepts of rheumatoid arthritis. *Nature* **423**, 356–361 (2003).
- Firestein, G. S. & McInnes, I. B. Immunopathogenesis of rheumatoid arthritis. *Immunity* **46**, 183–196 (2017).
- Burmester, G. R., Feist, E. & Doerner, T. Emerging cell and cytokine targets in rheumatoid arthritis. *Nat. Rev. Rheumatol.* **10**, 77–88 (2014).
- Goulielmos, G. N. et al. Genetic data: the new challenge of personalized medicine, insights for rheumatoid arthritis patients. *Gene* **583**, 90–101 (2016).
- McInnes, I. B. & Schett, G. Cytokines in the pathogenesis of rheumatoid arthritis. *Nat. Rev. Immunol.* **7**, 429–442 (2007).
- Noack, M. & Miossec, P. Selected cytokine pathways in rheumatoid arthritis. *Semin. Immunopathol.* **39**, 365–383 (2017).
- Choy, E. H. S. & Panayi, G. S. Mechanisms of disease: cytokine pathways and joint inflammation in rheumatoid arthritis. *New Engl. J. Med.* **344**, 907–916 (2001).
- Smolen, J. S. et al. New therapies for treatment of rheumatoid arthritis. *Lancet* **370**, 1861–1874 (2007).
- Bykerk, V. Unmet needs in rheumatoid arthritis. *J. Rheumatol.* **36**, 42–46 (2009).
- Taylor, P. C. et al. A structured literature review of the burden of illness and unmet needs in patients with rheumatoid arthritis: a current perspective. *Rheumatol. Int.* **36**, 685–695 (2016).
- Iwata, S. & Tanaka, Y. Progress in understanding the safety and efficacy of janus kinase inhibitors for treatment of rheumatoid arthritis. *Exp. Rev. Clin. Immunol.* **12**, 1047–1057 (2016).
- Smolen, J. S. et al. Treating rheumatoid arthritis to target: 2014 update of the recommendations of an international task force. *Ann. Rheum. Dis.* **75**, 3–15 (2016).
- Hu, C. M. J. et al. A biomimetic nanosponge that absorbs pore-forming toxins. *Nat. Nanotech.* **8**, 336–340 (2013).
- Hu, C. M. J. et al. Nanoparticle biointerfacing by platelet membrane cloaking. *Nature* **526**, 118–121 (2015).
- Copp, J. A. et al. Clearance of pathological antibodies using biomimetic nanoparticles. *Proc. Natl Acad. Sci. USA* **111**, 13481–13486 (2014).
- Thamphiwatana, S. et al. Macrophage-like nanoparticles concurrently absorbing endotoxins and proinflammatory cytokines for sepsis management. *Proc. Natl Acad. Sci. USA* **114**, 11488–11493 (2017).

18. Tak, P. P. et al. Analysis of the synovial cell infiltrate in early rheumatoid synovial tissue in relation to local disease activity. *Arthritis Rheum.* **40**, 217–225 (1997).
19. Horckmans, M. et al. Neutrophils orchestrate post-myocardial infarction healing by polarizing macrophages towards a reparative phenotype. *Eur. Heart J.* **38**, 187–197 (2017).
20. Dalli, J. et al. Heterogeneity in neutrophil microparticles reveals distinct proteome and functional properties. *Mol. Cell. Proteomics* **12**, 2205–2219 (2013).
21. Headland, S. E. Neutrophil-derived microvesicles enter cartilage and protect the joint in inflammatory arthritis. *Sci. Transl. Med.* **7**, 315ra190 (2015).
22. Wright, H. L., Moots, R. J. & Edwards, S. W. The multifactorial role of neutrophils in rheumatoid arthritis. *Nat. Rev. Rheumatol.* **10**, 593–601 (2014).
23. Wipke, B. T. & Allen, P. M. Essential role of neutrophils in the initiation and progression of a murine model of rheumatoid arthritis. *J. Immunol.* **167**, 1601–1608 (2001).
24. Cedergren, J., Forslund, T., Sundqvist, T. & Skogh, T. Intracellular oxidative activation in synovial fluid neutrophils from patients with rheumatoid arthritis but not from other arthritis patients. *J. Rheumatol.* **34**, 2162–2170 (2007).
25. Chakravarti, A., Raquil, M. A., Tessier, P. & Poubelle, P. E. Surface RANKL of Toll-like receptor 4-stimulated human neutrophils activates osteoclastic bone resorption. *Blood* **114**, 1633–1644 (2009).
26. Wright, H. L., Cox, T., Moots, R. J. & Edwards, S. W. Neutrophil biomarkers predict response to therapy with tumor necrosis factor inhibitors in rheumatoid arthritis. *J. Leukocyte Biol.* **101**, 785–795 (2017).
27. Hu, C.-M. J. et al. Erythrocyte membrane-camouflaged polymeric nanoparticles as a biomimetic delivery platform. *Proc. Natl Acad. Sci. USA* **108**, 10980–10985 (2011).
28. Declerck, L. S. et al. Expression of neutrophil activation markers and neutrophil adhesion to chondrocytes in rheumatoid-arthritis patients—relationship with disease-activity. *Res. Immunol.* **146**, 81–87 (1995).
29. Karsten, E., Breen, E. & Herbert, B. R. Red blood cells are dynamic reservoirs of cytokines. *Sci. Rep.* **8**, 3101 (2018).
30. Brzustewicz, E. et al. Heterogeneity of the cytokinome in undifferentiated arthritis progressing to rheumatoid arthritis and its change in the course of therapy. Move toward personalized medicine. *Cytokine* **97**, 1–13 (2017).
31. Ley, K., Laudanna, C., Cybulsky, M. I. & Nourshargh, S. Getting to the site of inflammation: the leukocyte adhesion cascade updated. *Nat. Rev. Immunol.* **7**, 678–689 (2007).
32. Schmidt, E. P. et al. The pulmonary endothelial glycocalyx regulates neutrophil adhesion and lung injury during experimental sepsis. *Nat. Med.* **18**, 1217–1223 (2012).
33. Brand, D. D., Latham, K. A. & Rosloniec, E. F. Collagen-induced arthritis. *Nat. Protoc.* **2**, 1269–1275 (2007).
34. Williams, R. O., Marinova-Mutafchieva, L., Feldmann, M. & Maini, R. N. Evaluation of TNF- α and IL-1 blockade in collagen-induced arthritis and comparison with combined anti-TNF- α /anti-CD4 therapy. *J. Immunol.* **165**, 7240–7245 (2000).
35. Joosten, L. A. B. et al. IL-1 α blockade prevents cartilage and bone destruction in murine type II collagen-induced arthritis, whereas TNF- α blockade only ameliorates joint inflammation. *J. Immunol.* **163**, 5049–5055 (1999).
36. Maia, M. et al. CD248 and its cytoplasmic domain a therapeutic target for arthritis. *Arthritis Rheum.* **62**, 3595–3606 (2010).
37. Pascual, V. et al. Role of interleukin-1 (IL-1) in the pathogenesis of systemic onset juvenile idiopathic arthritis and clinical response to IL-1 blockade. *J. Exp. Med.* **201**, 1479–1486 (2005).
38. Bartok, B. & Firestein, G. S. Fibroblast-like synoviocytes: key effector cells in rheumatoid arthritis. *Immunol. Rev.* **233**, 233–255 (2010).
39. Li, P. & Schwarz, E. M. The TNF- α transgenic mouse model of inflammatory arthritis. *Springer Semin. Immunopathol.* **25**, 19–33 (2003).
40. Shealy, D. J. et al. Anti-TNF- α antibody allows healing of joint damage in polyarthritic transgenic mice. *Arthritis Res.* **4**, R7 (2002).
41. Seeuws, S. et al. A multiparameter approach to monitor disease activity in collagen-induced arthritis. *Arthritis Res. Ther.* **12**, R160 (2010).
42. Kremer, J. M. et al. Treatment of rheumatoid arthritis by selective inhibition of T-cell activation with fusion protein ctla4ig. *New Engl. J. Med.* **349**, 1907–1915 (2003).
43. Kang, T. et al. Nanoparticles coated with neutrophil membranes can effectively treat cancer metastasis. *ACS Nano* **11**, 1397–1411 (2017).
44. Dolati, S. et al. Utilization of nanoparticle technology in rheumatoid arthritis treatment. *Biomed. Pharmacother.* **80**, 30–41 (2016).
45. Tabas, I. & Glass, C. K. Anti-inflammatory therapy in chronic disease: challenges and opportunities. *Science* **339**, 166–172 (2013).
46. Jie, Z. et al. Large-scale ex vivo generation of human neutrophils from cord blood CD34⁺ cells. *PLoS ONE* **12**, e0180832 (2017).
47. Feng, Q. et al. Scalable generation of universal platelets from human induced pluripotent stem cells. *Stem Cell Rep.* **3**, 817–831 (2014).

Acknowledgements

This work is supported by the Defense Threat Reduction Agency Joint Science and Technology Office for Chemical and Biological Defense under grant no. HDTRA1-14-1-0064 and National Science Foundation grant no. DMR-1505699.

Author contributions

Lia.Z. conceived and designed the experiments with W.G. and Q.Z. Q.Z., X.C., D.D., Y.Z., J.Z. and Lia.Z. performed all experiments. All authors analysed and discussed the data. Q.Z., W.G., R.H.F. and Lia.Z. wrote the paper.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information is available for this paper at <https://doi.org/10.1038/s41565-018-0254-4>.

Reprints and permissions information is available at www.nature.com/reprints.

Correspondence and requests for materials should be addressed to Lia.Z.

Publisher's note: Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Methods

Animal care. Mice were housed in an animal facility at the University of California San Diego (UCSD) under federal, state, local and National Institutes of Health (NIH) guidelines. All animal experiments were performed in accordance with NIH guidelines and approved by the Institutional Animal Care and Use Committee (IACUC) of UCSD.

Neutrophil collection. Fresh human peripheral blood neutrophils were purchased from ZenBio and activated within 16 h after blood collection from human donors. Neutrophils were purified by the manufacturer with OptiPrep density gradient centrifugation to a purity of ~95%. Briefly, leukocyte-rich plasma was layered over a 1.077/1.090 g ml⁻¹ OptiPrep gradient and centrifuged at 800g for 25 min. Neutrophil fraction was collected between a 1.077 and 1.090 g ml⁻¹ interface. Cells received were washed with 1× PBS and then suspended in serum-free RPMI media (Thermo Fisher Scientific) at a density of 2 × 10⁷ cells per ml. Neutrophils were then stimulated with 50 ng ml⁻¹ recombinant human TNF-α (Thermo Fisher Scientific) for 2 h at 37 °C. Stimulated cells were washed with 1× PBS, resuspended in 1:1 mixture of serum-free RPMI and HyCryo 2× cryopreservation medium (GE Healthcare), and stored at -80 °C for subsequent membrane derivation. Mouse neutrophils were collected from whole blood of ICR mice (six-week-old, male, Harlan Laboratories) using a modified Percoll gradient method. Specifically, before blood collection, 1.5 mg kg⁻¹ lipopolysaccharide (LPS) from *Escherichia coli* K12 (1 mg ml⁻¹ in 1× PBS, InvivoGen) was injected intraperitoneally into the mice to activate neutrophils in vivo. After 6 h, blood was collected by submandibular bleeding. Pooled blood was centrifuged (3,220g, 5 min, 4 °C) and the buffy coat on top was aspirated and placed over a three-layer Percoll gradient of 78, 69 and 52%. Samples were centrifuged (1,500g, 30 min, 4 °C) and the cell contents from the interface of the 69 and 78% gradient layers and the upper part of the 78% layer were collected. RBC lysis buffer (Thermo Fisher Scientific) was then added to the sample to lyse the erythrocytes. Neutrophils were purified by washing with 1× PBS three times, suspended, and stored at -80 °C for subsequent membrane derivation.

Mouse neutrophils were co-stained with fluorescein isothiocyanate (FITC) anti-mouse Gr-1 antibody to identify neutrophil cells and PE anti-mouse LFA-1 antibody (Biolegend) to confirm neutrophil activation. A series of experiments were conducted to optimize the neutrophil isolation process. In the collection time optimization experiment, mice were injected with 1.5 mg kg⁻¹ LPS and whole blood was collected at the designated time points. In the LPS dosage optimization experiment, mice were injected with the designated dosage of LPS, and whole blood was collected at 6 h post-injection. Washed mouse neutrophils were resuspended in PBS at 1 × 10⁶ cells per ml and analysed by a Becton Dickinson FACSCanto-II flow cytometer. Results were analysed using FlowJo software.

Cell membrane derivation. Plasma membrane of neutrophils was harvested following a previously published protocol⁴⁹. Briefly, frozen cells were thawed and washed with 1× PBS three times (centrifugation at 800g). Cells were then suspended in hypotonic lysing buffer containing 30 mM Tris-HCl (pH 7.5), 225 mM D-mannitol, 75 mM sucrose, 0.2 mM EGTA (all reagents from Sigma), and a protease and phosphatase inhibitor cocktail (Thermo Fisher Scientific). Cells were then disrupted using a Dounce homogenizer with a tight-fitting pestle (20 passes). The homogenized solution was centrifuged at 20,000g for 25 min at 4 °C. The pellet was discarded and the supernatant was centrifuged again at 100,000g for 35 min at 4 °C. Following the centrifugation, membranes were collected as the pellet and washed twice with 0.2 mM EDTA in water. Membrane content was quantified using a BCA kit (Pierce) in reference to a bovine serum albumin (BSA) standard. Approximately 125 million human neutrophils or 200 million mouse neutrophils were able to yield 1 mg membrane material (protein weight). Membrane was suspended with 0.2 mM EDTA to a protein concentration of 4 mg ml⁻¹ and stored at -80 °C for subsequent studies.

To derive RBC membrane, washed RBCs were resuspended in 0.25× PBS on ice for 20 min, then centrifuged at 800g for 5 min. The haemoglobin was removed, and the pink pellet was resuspended in 0.25× PBS on ice for 20 min. The process was repeated until the haemoglobin was completely removed. The RBC membrane was resuspended in water at a protein concentration of 4 mg ml⁻¹ and stored at -80 °C for subsequent studies.

Synthesis of nanoparticles. Neutrophil-NPs were synthesized using a sonication method. Briefly, to synthesize nanoparticle cores, 0.2 ml poly(DL-lactic-co-glycolic acid) (50:50 PLGA, 0.67 dl g⁻¹, Lactel Absorbable Polymers) in acetone (20 mg ml⁻¹) was added dropwise into 1 ml water. The solution was placed under a vacuum aspirator until the acetone had evaporated completely. For fluorescence imaging experiments, 1,1'-dioctadecyl-3,3,3',3'-tetramethylindodicarbocyanine perchlorate (DiD, excitation = 644 nm/emission = 665 nm, Thermo Fisher Scientific) was encapsulated into PLGA cores (0.1 wt%). For membrane coating, neutrophil membrane was mixed with PLGA cores at a polymer-to-membrane protein weight ratio of 2:1. The mixture was then sonicated with a bath sonicator for 2 min (Fisher Scientific FS30D). RBC-NPs were synthesized following the same procedure. Coating completeness was confirmed by studying nanoparticle stability in 1× PBS⁵⁰.

Nanoparticle characterization. Neutrophil-NPs were measured for hydrodynamic size and surface zeta potential with DLS (ZEN 3600 Zetasizer, Malvern). To examine

the morphology, nanoparticles were stained with uranyl acetate (0.2 wt%) and visualized using transmission electron spectroscopy (FEI 200kV Sphera). Neutrophil-specific surface markers on neutrophils, neutrophil membranes and neutrophil-NPs were examined by western blotting. In the study, all samples were prepared in lithium dodecyl sulfate (LDS) loading buffer and run at equivalent protein concentrations on NuPAGE Novex 4–12% Bis-Tris minigels in MOPS running buffer. The gel was transferred onto a nitrocellulose membrane and probed with antibodies specific for human or mouse CD120b, IL-1R and LFA-1, along with horseradish peroxidase-conjugated secondaries against mouse IgG or rat IgG (Biolegend).

Cell culture. Primary human chondrocytes between passages 2 and 6 were cultured in human chondrocyte growth medium (both from Cell Applications) at 37 °C in a 5% CO₂ environment. HUVECs (ATCC) were maintained in human endothelial cell growth medium (Cell Applications) at 37 °C in a 5% CO₂ environment. Cell lines were purchased with certification of authentication and free from Mycoplasma.

Neutrophil-NP adhesion assay. HUVECs or human chondrocytes were seeded in 12-well tissue culture plates at 50% confluency and cultured overnight. Cell culture medium was changed and recombinant human TNF-α or IL-1β (Thermo Fisher Scientific) was added to the desired concentrations. After 6 h of stimulation, cells were washed with 1× PBS and fixed with 10% phosphate buffered formalin (Fisher Scientific) for 10 min and blocked with 1% BSA for 1 h. The cells were then incubated with 0.2 mg ml⁻¹ DiD-labelled neutrophil-NPs or RBC-NPs in PBS at 4 °C for 60 s. After incubation, cells were washed five times with ice-cold PBS, mounted with Vectashield Antifade Mounting Medium with 4,6-diamidino-2-phenylindole (DAPI; Vector Laboratories), and imaged with an EVOS inverted fluorescence microscope (Thermo Fisher Scientific). For flow cytometric analysis, cells were scraped and collected after PBS wash, then analysed with a Becton Dickinson FACSCanto-II flow cytometer. Results were analysed using FlowJo software.

Quantification of cytokine binding. Recombinant human TNF-α (Thermo Fisher Scientific, final concentration 8.82 ± 0.80 ng ml⁻¹) or human IL-1β (Thermo Fisher Scientific, final concentration 7.92 ± 1.28 ng ml⁻¹) was mixed with neutrophil-NPs at final concentrations ranging from 0 to 4 mg ml⁻¹. The mixtures were incubated for 2 h at 37 °C and then centrifuged at 16,100g for 10 min to remove the nanoparticles. Nanoparticle removal was confirmed by comparing the fluorescence signal from DiD-labelled neutrophil-NP suspension and supernatant after nanoparticles were centrifuged at 16,100g for 10 min. Cytokine concentration in the supernatant was quantified by human TNF-α or human IL-1β enzyme-linked immunosorbent assay (ELISA) kits (Biolegend). Nonlinear fitting of curves was performed in GraphPad Prism 7.

Chondrocyte apoptosis assay. Human chondrocytes were seeded in 12-well tissue culture plates at a density of 1 × 10⁶ cells per well and cultured overnight. TNF-α or IL-1β was incubated with neutrophil-NPs or RBC-NPs in chondrocyte culture medium at the desired concentrations at 37 °C for 2 h. Human synovial fluid (hSF) samples from rheumatoid arthritis patients (BioIVT) were filtered with a sterile 0.22 μm syringe filter (Spectrum Laboratories) and incubated with neutrophil-NPs or RBC-NPs in chondrocyte culture medium at 37 °C for 2 h. The mixture was then centrifuged at 16,100g for 10 min to remove the nanoparticles. The supernatant was used to stimulate chondrocytes for 48 h. The cells were detached with 0.25% trypsin-EDTA (Thermo Fisher Scientific) and washed with sterile PBS. To determine mitochondrial activity, cells were stained with DiOC₆ (100 nM final concentration from 10 mM stock in 100% ethanol, Life Technologies) for 30 min at 37 °C. The cells were washed with PBS and analysed using a Becton Dickinson FACSCanto-II flow cytometer. Results were analysed using FlowJo software. For fluorescent imaging experiments, a chondrocyte monolayer was cultured, stimulated and imaged with an EVOS inverted fluorescence microscope. Apoptotic cells are FITC^{dim} and non-apoptotic cells are FITC^{bright}. In hSF experiments, hSF samples (BioIVT) were incubated with neutrophil-NPs or RBC-NPs under the same condition. Supernatant was used to stimulate the cells as described above. Statistical analysis was performed using paired *t*-test (*n* ≥ 3).

Chondrocyte activation assay. Chondrocytes were prepared as described above. Cytokines were incubated with nanoparticles in culture medium at 37 °C for 2 h. Filtered hSF samples from rheumatoid arthritis patients were incubated with neutrophil-NPs or RBC-NPs in chondrocyte culture medium at 37 °C for 2 h. The mixture was then centrifuged at 16,100g for 10 min to remove the nanoparticles. The supernatant was used to stimulate chondrocytes for 6 h. Cells were then washed, fixed and blocked. Cells were stained with anti-human CD54 (Biolegend) for 30 min at room temperature, then stained with FITC-anti-mouse IgG (Biolegend). Washed cells were scraped and analysed by a Becton Dickinson FACSCanto-II flow cytometer. Results were analysed using FlowJo software. For fluorescent imaging experiments, chondrocyte and HUVEC monolayers were cultured and stimulated in tissue culture plates. ICAM-1 expression was probed with anti-human CD54 antibody as described above, then detected with Alexa647-anti-mouse IgG (for HUVECs, Biolegend) and FITC-anti-mouse IgG (for chondrocytes). Cells were mounted with Vectashield Antifade Mounting Medium

with DAPI. An EVOS inverted fluorescence microscope was used to image the cells. In hSF experiments, hSF samples (BioIVT) were incubated with neutrophil-NPs or RBC-NPs under the same condition. Supernatant was used to stimulate the cells as described above. Statistical analysis was performed using paired *t*-test ($n \geq 3$).

Chondrocyte MMP-3 secretion assay. Chondrocytes were seeded onto 96-well tissue culture plates (1×10^5 cells per well) and cultured overnight. Filtered hSF samples were diluted 4× with chondrocyte culture medium and added to chondrocytes. Neutrophil-NPs or RBC-NPs were added to chondrocytes to achieve a final concentration of 2 mg ml^{-1} . At designated time points, MMP-3 concentration from undiluted culture medium was quantified with a human MMP-3 PicoKine ELISA kit (Boster Biological Technology). Statistical analysis was performed using repeated-measure one-way ANOVA ($n \geq 3$).

Chondrocyte aggrecan production assay. Cells were prepared in the same way as described above. Filtered hSF samples were diluted 4× with chondrocyte culture medium and added to the chondrocytes. Neutrophil-NPs or RBC-NPs were added to chondrocytes to achieve a final concentration of 2 mg ml^{-1} . Cells were cultured at 37°C for 24 h. The culture medium was then replaced with fresh medium, and fresh hSF samples and nanoparticles were added. This treatment was repeated three times. Then chondrocytes were washed and stained with anti-human aggrecan (Biolegend) at room temperature for 2 h, and further probed with FITC-anti-mouse IgG at room temperature for 30 min. Washed cells were analysed by a Becton Dickinson FACSCanto-II flow cytometer. Results were analysed using FlowJo software. For fluorescence imaging experiment, cells were mounted with Vectashield Antifade Mounting Medium with DAPI and imaged with an EVOS inverted fluorescence microscope. Statistical analysis was performed using paired *t*-test ($n \geq 3$).

Neutrophil-NP penetration study. Mouse femoral heads were collected from 12-week-old ICR mice and cultured for 48 h in high glucose serum-free DMEM medium (4.5 g l^{-1} glucose, Hyclone). Following the culture, 10 ng ml^{-1} of recombinant mouse IL-1 β (ThermoFisher Scientific) was added to the culture on days 0, 1 and 2. Culture medium was changed each time before IL-1 β was added. On day 3, 0.2 mg ml^{-1} of DiD-labelled mouse neutrophil-NPs were incubated with the femoral heads at 37°C for 24 h. After incubation, femoral heads were washed with PBS and embedded in Tissue-Tek OCT Compound (Sakura Finetek) for frozen sectioning. Frozen sections were mounted (Vectashield Antifade Mounting Medium with DAPI) and imaged with an EVOS inverted fluorescence microscope. Images were analysed with ImageJ to quantify nanoparticle penetration into the cartilage. Specifically, images were divided into 5- μm -thick sections, starting from the distal femoral head surface. Fluorescence within each image section was measured with ImageJ and normalized to the fluorescence of the outermost tissue of the neutrophil-NP-treated femoral head. Mouse RBC-NPs were used as the control.

Mouse femoral head degradation assay. Mouse femoral heads were collected using the same procedure as described above. To study degradation, 10 ng ml^{-1} of IL-1 β was mixed with 2 mg ml^{-1} neutrophil-NPs in medium and added to femoral head cultures to stimulate cartilage degradation for 48 h. The medium was then changed, and the same amount of cytokine-nanoparticle mixture was added to the femoral heads. A total of three treatments, 48 h each, were applied. The samples were then fixed and processed for safranin-O staining, and the safranin-O-positive area was quantified with ImageJ. PBS and mouse RBC-NP-treated samples served as controls.

Mouse models of inflammatory arthritis. A CIA mouse model was established by following a previously published protocol with immunizations on days 1 and 21 (ref. ³⁵). Human TNF- α transgenic mice (5-week-old male) were obtained from Taconic Biosciences. Mice were randomized before the experiment.

In vivo efficacy study protocols. To study therapeutic efficacy with CIA mice, $20 \mu\text{l}$ of mouse neutrophil-NPs (2 mg ml^{-1}) was injected into each knee joint of the hind legs on days 28 (first day of obvious clinical signs of arthritis), 30, 32, 34, 36 and 38. Sterile PBS as a negative control was injected intra-articularly to mice on the same days. Anti-mouse IL-1 β (3 mg kg^{-1}) or an anti-mouse TNF- α

(3 mg kg^{-1}) were used as positive controls and injected intraperitoneally to mice three times a week from day 28 to 60. To study therapeutic efficacy with human TNF- α transgenic mice, $20 \mu\text{l}$ of mouse neutrophil-NPs (2 mg ml^{-1}) was injected into each knee joint of the hind legs on days 42, 45, 49, 52, 56, 59, 63 and 66. Sterile PBS (negative control) was injected intra-articularly on the same days. Anti-human TNF- α (positive control, 10 mg kg^{-1} , Biolegend) was administered intraperitoneally on days 42, 49, 56 and 63. To study prophylactic efficacy with CIA mice, $20 \mu\text{l}$ of mouse neutrophil-NPs (2 mg ml^{-1}) was injected into each knee joint of the hind legs on days 7, 14 and 21. Sterile PBS was injected to control mice intra-articularly on the same days. Positive control mice were injected intraperitoneally with an anti-mouse IL-1 β (3 mg kg^{-1}) or an anti-mouse TNF- α (3 mg kg^{-1}) three times a week from day 7 to day 21.

Histological analysis of mouse knee joints. At study endpoints, mice were euthanized and hind knee joints were collected for H&E staining or safranin-O staining. To analyse fibroblast-like synoviocyte activation, knee joint sections were stained with a rabbit anti-CD248 (Bioss Antibodies) or a rabbit anti-fibronectin (Novus Biologicals). Biotinylated anti-rabbit IgG was used as the secondary antibody for chromagen development. Sections were counter-stained with haematoxylin to visualize cell nuclei. Images were taken with a Micromaster II Microscope (Fisher Scientific). The safranin-O-positive area was quantified by using ImageJ.

Quantification of serum cytokines. CIA mouse serum samples were collected on days 8, 15, 22, 29, 36, 43, 50 and 57 and concentrations of IL-1 β and TNF- α were quantified with ELISA. Specifically, the whole blood of CIA mice was collected by submandibular bleeding into microtubes and allowed to clot at room temperature for 30 min. Samples were then centrifuged at $2,000g$ for 6 min to collect serum from the supernatant. Serum samples were immediately frozen at -20°C until analysis by using mouse IL-1 β and mouse TNF- α ELISA kits (Biolegend) within 3 days of collection.

Joint swelling assessment and arthritis scoring. Mouse hind knee joint diameters were measured with a digital caliper at study endpoints. Images of mouse paws were taken every 2 days from day 0 to day 60, and paw volumes were calculated by analysing images with ImageJ. Specifically, the length, width and thickness of mouse paws were measured from the top- and side-view images of each paw. The volume was approximated as the product of these three parameters assuming equivalent importance of each in reflecting the extent of inflammation in the paw. Paw scorings (score ~0–4) were given by a blinded researcher based on the following criteria: 0, normal; 1, mild redness of ankle or tarsal joints; 2, mild redness and swelling extending from ankle to the tarsals; 3, moderate redness and swelling from ankle to metatarsal joints; 4, severe redness and swelling encompassing the ankle, foot and digits. Hind ankle joint diameters were measured with a digital caliper at day 60.

Statistical analysis was performed using one-way ANOVA. Paw scoring data were analysed using the Kruskal–Wallis test. Replicates represent different mice subjected to the same treatment. In CIA studies, $n = 7$ for all groups. In the human TNF- α transgenic mouse study, $n = 5$ for the neutrophil-NP group and the anti-TNF- α group, and $n = 3$ for the PBS group.

Data availability

The data that support the plots within this paper and other findings of this study are available from the corresponding author upon reasonable request.

References

- Graham, J. M. Isolation of human polymorphonuclear leukocytes (granulocytes) from a leukocyte-rich fraction. *Sci. World J.* **2**, 1393–1396 (2002).
- Fang, R. et al. Cancer cell membrane-coated nanoparticles for anticancer vaccination and drug delivery. *Nano. Lett.* **14**, 2181–2188 (2014).
- Luk, B. T. et al. Interfacial interactions between natural RBC membranes and synthetic polymeric nanoparticles. *Nanoscale* **6**, 2730–2737 (2014).

Reporting Summary

Nature Research wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Research policies, see [Authors & Referees](#) and the [Editorial Policy Checklist](#).

Statistical parameters

When statistical analyses are reported, confirm that the following items are present in the relevant location (e.g. figure legend, table legend, main text, or Methods section).

n/a Confirmed

- ☐ ☒ The exact sample size (n) for each experimental group/condition, given as a discrete number and unit of measurement
- ☐ ☒ An indication of whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
- ☐ ☒ The statistical test(s) used AND whether they are one- or two-sided
Only common tests should be described solely by name; describe more complex techniques in the Methods section.
- ☐ ☒ A description of all covariates tested
- ☐ ☒ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
- ☐ ☒ A full description of the statistics including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
- ☒ ☐ For null hypothesis testing, the test statistic (e.g. F , t , r) with confidence intervals, effect sizes, degrees of freedom and P value noted
Give P values as exact values whenever suitable.
- ☒ ☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
- ☒ ☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
- ☒ ☐ Estimates of effect sizes (e.g. Cohen's d , Pearson's r), indicating how they were calculated
- ☐ ☒ Clearly defined error bars
State explicitly what error bars represent (e.g. SD, SE, CI)

Our web collection on [statistics for biologists](#) may be useful.

Software and code

Policy information about [availability of computer code](#)

Data collection

No stand-alone or custom software was used.

Data analysis

Softwares used in analysis include FlowJo VX, GraphPad Prism 6, ImageJ, Microsoft Excel.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors/reviewers upon request. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Research [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A list of figures that have associated raw data
- A description of any restrictions on data availability

The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Field-specific reporting

Please select the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see [nature.com/authors/policies/ReportingSummary-flat.pdf](https://www.nature.com/authors/policies/ReportingSummary-flat.pdf)

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	The in vitro experiments described in this manuscript were performed 3 times. Standard number of replication is 3 in all in vitro studies. A minimum of 3 independent repeats is required to evaluate significance. The in vivo efficacy studies were performed with 7 mice per group for non-transgenic animals, 3-5 for transgenic animals (5 for neutrophil-NP and anti-TNF- α antibody groups, 3 for PBS group), per standard protocols and to minimize the number of animals under experiment. Details regarding sample size of all experiments are provided in the Methods section and figure legends.
Data exclusions	No data were excluded.
Replication	Experiments were repeated and experimental findings were reproducible.
Randomization	Samples were randomly allocated to corresponding experimental groups. Organisms were cultured and maintained in the same environment and randomly allocated to each group.
Blinding	Investigators were not blinded.

Reporting for specific materials, systems and methods

Materials & experimental systems

n/a	Involved in the study
☒	☐ Unique biological materials
☐	☒ Antibodies
☐	☒ Eukaryotic cell lines
☒	☐ Palaeontology
☐	☒ Animals and other organisms
☒	☐ Human research participants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☐	☒ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used	FITC anti-mouse Gr-1 (Cat#108405; Clone#RB6-8C5), PE anti-mouse LFA-1 (Cat#141005; Clone#H155-78), anti-human CD120b (Cat#358402; Clone#3G7A02), anti-human LFA-1 (Cat#363402; Clone#m24), anti-human CD54 (Cat#353102; Clone#HA58), anti-mouse CD120b (Cat#113302; Clone#TR75-54.7), anti-mouse IL-1R (Cat#113502; Clone#JAMA-147), anti-mouse LFA-1 (Cat#141002; Clone#H155-78), FITC anti-mouse IgG (Cat#406001; Clone#405305), Alexa647 Goat anti-mouse IgG (Cat#405322; Clone#Poly4053), HRP Goat anti-mouse IgG (Cat#405306; Clone#Poly4053), HRP Goat anti-rat IgG (Cat#405405; Clone#Poly4054) were purchased from Biolegend. Rabbit anti-CD248 antibody (bs-2101R) was purchased from Bioss Antibodies. Rabbit anti-fibronectin (NBP1-91258) antibody was purchased from Novus Biologicals. HRP anti-Armenian hamster IgG (Cat#6060-05) was purchased from and verified by SouthernBiotech (www.southernbiotech.com). Anti-NITEGE (aggrecanase-neoepitope, MAB19310) antibody was purchased from Milipore Sigma. Anti-FFGVG (MMP-neoepitope, NB100-74350) antibody was purchased from Novus Biologicals.
Validation	All antibodies were verified by the supplier and each lot has been quality tested.

Eukaryotic cell lines

Policy information about [cell lines](#)

Cell line source(s)	Human chondrocytes (Cell Applications, 402-05a) were isolated from healthy human cartilage explants by the supplier and cultured in chondrocyte growth medium (Cell Applications, 411-500). Human umbilical vein endothelial cells (HUVECs) were
---------------------	--

	purchased from ATCC (CRL-1730) and cultured in endothelial growth medium (Cell Applications, 211-500).
Authentication	Cells were used without modification once received from the supplier and therefore were not authenticated.
Mycoplasma contamination	All cell lines were tested negative for Mycoplasma contamination per suppliers.
Commonly misidentified lines (See ICLAC register)	n/a

Animals and other organisms

Policy information about [studies involving animals](#); [ARRIVE guidelines](#) recommended for reporting animal research

Laboratory animals	Mouse neutrophils were isolated from whole blood of LPS-stimulated ICR mice (Harlan Laboratory, male, 6-weeks-old, 22-24g) and used without further modification. Mouse femoral head cartilage explants were isolated from healthy ICR mice (Harlan Laboratory, male, 6-weeks-old, 22-24g). DBA/1J Mice in collagen-induced arthritis study (Jackson Laboratory, male, Stock#000670, 6-weeks-old, 18-22g) were used without special care. Human TNF- α transgenic mice (5-week-old male) were obtained from Taconic Biosciences and allowed to accustom to standard housing conditions for one week. Mice were randomized before the experiment.
Wild animals	n/a
Field-collected samples	n/a

Flow Cytometry

Plots

Confirm that:

- ☒ The axis labels state the marker and fluorochrome used (e.g. CD4-FITC).
- ☒ The axis scales are clearly visible. Include numbers along axes only for bottom left plot of group (a 'group' is an analysis of identical markers).
- ☒ All plots are contour plots with outliers or pseudocolor plots.
- ☒ A numerical value for number of cells or percentage (with statistics) is provided.

Methodology

Sample preparation	Human neutrophils were purchased from Zen-Bio (SER-CUSTOM). Mouse neutrophils were isolated from whole blood of LPS-stimulated ICR mice (Harlan Laboratory) by Percoll density gradient centrifugation. Human chondrocytes were isolated from healthy human cartilage explants (Cell Applications, 402-05a) and cultured in chondrocyte growth media (Cell Applications, 411-500). Human umbilical vein endothelial cells (HUVECs) were purchased from ATCC (CRL-1730) and cultured in endothelial growth media (Cell Applications, 211-500). Cells were used without further modification. Details on immunofluorescence staining are provided in Methods.
Instrument	Becton Dickinson FACSCanto-II flow cytometer.
Software	FlowJo VX
Cell population abundance	n/a
Gating strategy	For mouse neutrophil characterization, preliminary FSC/SSC gates were determined by comparing the population of control sample (mouse RBCs) with positive mouse neutrophil sample. LFA-1 staining boundaries were clearly defined in the resulting neutrophil population (as shown in Supplementary Information Figure 5-7). For in vitro studies, preliminary FSC/SSC gates were determined by naive chondrocytes or HUVECs samples. Staining boundaries were determined by comparing the fluorescence signals of naive cells with cytokine-activated cells.

☐ Tick this box to confirm that a figure exemplifying the gating strategy is provided in the Supplementary Information.